Homework 15.10.'26

↳ Exercises 2.2 and 2.3

2.2 Show that $\mathcal{C} = \delta(\mathcal{C})$ if and only if $\mathcal{C}$ is a sigma algebra.

$\delta(\mathcal{C})$ ... smallest(!) δ-algebra, generated by $\mathcal{C}$

1. $\mathcal{C}$ is δ-algebra $\Rightarrow$ $\mathcal{C} = \delta(\mathcal{C})$ follows from definition of $\delta(\mathcal{C})$ □
2. $\mathcal{C} = \delta(\mathcal{C}) \Rightarrow \mathcal{C}$ is δ-algebra

   well if $\mathcal{C}$ is equal to a δ-alg, it is a δ-alg ?!? → not enough, use $\delta() = \cap \bar{\mathcal{F}}$ ...

   ↳ forgot $\delta(\mathcal{C})$ is smallest

   Let $\mathcal{F}_i$, $i \in \mathbb{N}$ be all δ-algebras that contain $\mathcal{C}$, so that $\delta(\mathcal{C}) = \bigcap_{i=1}^{\infty} \mathcal{F}_i$ and presuppose $\mathcal{C} = \delta(\mathcal{C}) = \bigcap_{i=1}^{\infty} \mathcal{F}_i$

   We know (from practise sessions) that an intersection of δ-alg. is a δ-alg, therefore $\mathcal{C}$ is.

2.3 Let $\mathcal{C}$ and $\mathcal{D}$ be two collections of subsets on Ω s.t. $\mathcal{C} \subset \mathcal{D}$. Prove that $\delta(\mathcal{C}) \subseteq \delta(\mathcal{D})$

$\delta(\mathcal{C}) \subseteq \delta(\mathcal{D}) \sim \neg\, \delta(\mathcal{C}) \supset \delta(\mathcal{D})$ → proof by contradiction?

Let's suppose $\mathcal{C} \subseteq 2^{\Omega}$, $\mathcal{D} \subseteq 2^{\Omega}$, $\mathcal{C} \subset \mathcal{D}$ and $\delta(\mathcal{C}) \supset \delta(\mathcal{D})$ // scratch that

We have $\mathcal{C} \subset \mathcal{D} \subseteq 2^{\Omega}$. Let us prove that any element of $\delta(\mathcal{C})$ is also element of $\delta(\mathcal{D})$

$\delta(\mathcal{D}) = \bigcap_{i=1}^{\infty} \mathcal{F}_i$, where $\mathcal{F}_i$ are δ-alg. s.t. $\mathcal{D} \subseteq \mathcal{F}_i$ $\forall i$

since $\mathcal{C} \subset \mathcal{D}$ and $\mathcal{D} \subseteq \mathcal{F}_i$ $\Rightarrow \mathcal{C} \subseteq \mathcal{F}_i$, which means $\delta(\mathcal{C}) \subseteq \delta(\mathcal{D})$

... too little?

// more elegant with $\delta(\mathcal{C})$ ↓

$\delta(\mathcal{C}) = \bigcap_{i=1}^{\infty} \mathcal{F}_i$ where $\mathcal{F}_i$ are δ-alg. that contain $\mathcal{C}$. $\delta(\mathcal{D})$ contains $\mathcal{D}$, is a δ-alg. and $\mathcal{C} \subset \mathcal{D}$, therefore $\exists j$ $\mathcal{F}_j = \delta(\mathcal{D})$ Since $A \cap B \subseteq B$ it follows that $\bigcap_{i=1}^{\infty} \mathcal{F}_i \subseteq \mathcal{F}_j = \delta(\mathcal{D})$ □